Binary Measure, Arithmetic Half-Frequency, and Spectral Statistics

Finite Dyadic Pushforwards, Deterministic Sign Records, and Zeta-Zero Comparisons

Christopher M Struck

ORCID: 0009-0009-4291-6782

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Abstract

The Bare Binary Biconditional connected exchange-invariant binary probability, the fair Bernoulli product law, almost-sure half-frequency, and Lebesgue measure under dyadic coding. This paper completes the three-study half-probability-convergence program related to that construction while keeping the studies' logical roles separate. The first is a direct finite-depth image of the exact measure identity: 400,000 depth-40 binary records sampled with parameter $p = 1/2$ give a histogram consistent with uniform measure on $[0, 1]$, while $p = 0.35$ gives the nonuniform finite-resolution pattern of a measure known analytically to be singular with respect to Lebesgue measure. The second examines three deterministic arithmetic binary records through an arithmetic cutoff of $3{,}000{,}000$: the Liouville sign, the nonzero Möbius sign, and the residue class modulo 4 of odd primes. Their observed head frequencies are respectively 0.499843, 0.500029, and 0.499426, but their eligible-trial counts, exact ties, and sign-change textures differ substantially. These deterministic records are not modeled as independent Bernoulli trials. The third reproduces the established Montgomery–Odlyzko spectral-statistics comparison using 1,200 cached nontrivial zeta zeros, with 1,190 zeros retained after a low-height cutoff and 1,189 nearest-neighbor spacings. It is included as a calibration and as wider deterministic-random context, not as evidence for the binary measure theorem. Together, the three studies provide a reproducible illustrated analysis of the earlier synthesis without claiming to prove it computationally, establish asymptotic rates from finite data, or derive a number-theoretic conclusion. The separate question of observation-record convergence is addressed independently.

1 Relation to the Bare Binary Biconditional

The result in [1] states, within the Bernoulli family, the exact chain

$$\tau_*\nu_p = \nu_p \quad \Longleftrightarrow \quad p = \frac{1}{2} \quad \Longleftrightarrow \quad \phi_*\mu_p = \lambda \quad \Longleftrightarrow \quad \mu_p\left(\left\{\omega : \lim_{n\to\infty} \frac{S_n(\omega)}{n} = \frac{1}{2}\right\}\right) = 1,$$

where ν_p is the binary law with $\nu_p(\{1\}) = p$, τ exchanges 0 and 1, $\mu_p = \nu_p^{\otimes\mathbb{N}}$, and

$$\phi(\omega) = \sum_{k=1}^{\infty} \omega_k 2^{-k}.$$

That chain is a theorem-level statement assembled from established measure and probability results. It does not need numerical confirmation. The purpose of this paper is different: it presents finite computations that display selected components and comparisons, identifies which plots directly illustrate an analytic identity

and which serve only as contextual comparisons, and states enough methodology for every reported number and figure to be regenerated.

The Bare Binary Biconditional was formulated before the half-probability-convergence program presented here was proposed. From a broader set of proposed research options, the author selected the deterministic arithmetic-sign and zeta-zero studies for that program because of their relationship to the pre-existing biconditional. The established results assembled in [1] supported the biconditional without requiring completion of those studies, so the synthesis paper superseded the half-probability-convergence program as the first completed output. The dyadic-pushforward study was subsequently added from [1] as a direct computational representation of its exact measure identity, and the present paper completes the resulting three-study half-probability-convergence program.

The three computations have different logical relationships to the result in [1]:

Table 1: Logical roles of the three computational studies.

Object	**Method**	**Interpretive role**
Bernoulli records under the dyadic map	Finite Monte Carlo sampling	Direct finite-depth image of the exact pushforward statement
Deterministic arithmetic signs	Exact finite enumeration	Comparison with half-frequency and square-root-scale behavior; no Bernoulli assumption
Unfolded zeta-zero spacings and pair correlation	Finite spectral-statistics calculation	Reproduction and calibration in the established deterministic-spectral context

This separation is deliberate. It prevents a visual resemblance in the deterministic arithmetic or spectral calculations from being used as a premise in the exact dyadic argument.

2 Computational Methodology and Claim Boundaries

The dyadic calculation draws 400,000 independent binary words of length 40 for each of $p = 1/2$ and $p = 0.35$, using a pseudorandom generator with fixed seed `20260827`. Each word is mapped to its truncated dyadic coordinate $\phi_{40}(\omega) = \sum_{k=1}^{40} \omega_k 2^{-k}$, and each resulting sample is summarized in 256 equal-width bins on $[0, 1]$.

The arithmetic calculations are deterministic. The Liouville and Möbius values are reconstructed through $n \leq 3{,}000{,}000$ with a smallest-prime-factor sieve, and odd primes are identified by a sieve of Eratosthenes before their residue classes modulo 4 are encoded as signs. For each eligible record, the calculation reports the number of trials, the empirical proportion of $+1$, the terminal running sum, exact zero ties, and sign changes after zero ties are omitted. A second comparison truncates all three records at the common horizon of 216,815 eligible observations.

The spectral calculation begins with a stored set of the first 1,200 positive nontrivial zeta-zero ordinates. Ordinates at or below 50 are omitted, leaving 1,190 values. The retained ordinates are unfolded with the smooth Riemann–von Mangoldt term; the resulting 1,189 nearest-neighbor spacings are histogrammed and compared with the GUE Wigner surmise and the Poisson density. Pairwise unfolded differences up to three mean spacings are summarized in 60 bins and compared with Montgomery's limiting curve.

The computations use Python 3.10, NumPy, Matplotlib, and, for generation of the stored zeta-zero data, mpmath. The complete scripts, cached zero data, and frozen figure files remain in the accompanying research archive. Randomized calculations use the fixed seed stated above; the arithmetic and spectral calculations use fixed input data and deterministic transformations.

Throughout this paper:

- *exact* refers to an analytic identity already established independently of the computation;
- *observed* refers to the finite data generated by the stated methodology;
- *benchmark* refers to a comparison curve, not an estimated asymptotic law; and
- *illustrates* does not mean proves, confirms, or physically explains.

3 Finite-Depth Dyadic Pushforwards

3.1 Exact statement

For (0<p<1), let μ_p be the Bernoulli product measure on $\{0,1\}^{\mathbb{N}}$. At finite depth d, a word $a_1 \dots a_d$ has cylinder probability

$$\mu_p([a_1 \dots a_d]) = p^{a_1+\cdots+a_d}(1-p)^{d-(a_1+\cdots+a_d)}.$$

The dyadic map sends that word to a dyadic interval of length 2^{-d}. When $p = 1/2$, every depth-d cylinder has mass 2^{-d}, exactly matching the length of its image interval. Agreement on all dyadic cylinders yields

$$\phi_* \mu_{1/2} = \lambda.$$

For $p \neq 1/2$, the Bernoulli law gives digit frequency p almost surely, whereas Lebesgue-almost every binary expansion has digit frequency $1/2$ [2,3]. The two measures therefore concentrate on disjoint typical-frequency sets, so

$$\phi_* \mu_p \perp \lambda \qquad (p \neq 1/2).$$

For $0 < p < 1$, the biased pushforward still has full topological support: every nonempty dyadic interval has positive probability. Singularity does not mean that intervals outside a Cantor set receive zero mass. It means that a Borel set can have full pushforward measure and zero Lebesgue measure.

3.2 Finite computation

The simulation draws two arrays of 400,000 binary words, each of depth 40. One array uses $p = 1/2$; the other uses $p = 0.35$. Each word is mapped by the truncated sum

$$\phi_{40}(\omega) = \sum_{k=1}^{40} \omega_k 2^{-k}.$$

At the 256-bin resolution of the plot, the omitted dyadic tail is much smaller than a bin. The fair sample has mean 0.49981, compared with its exact expectation $1/2$. The biased sample has mean 0.34980, compared with its exact expectation 0.35.

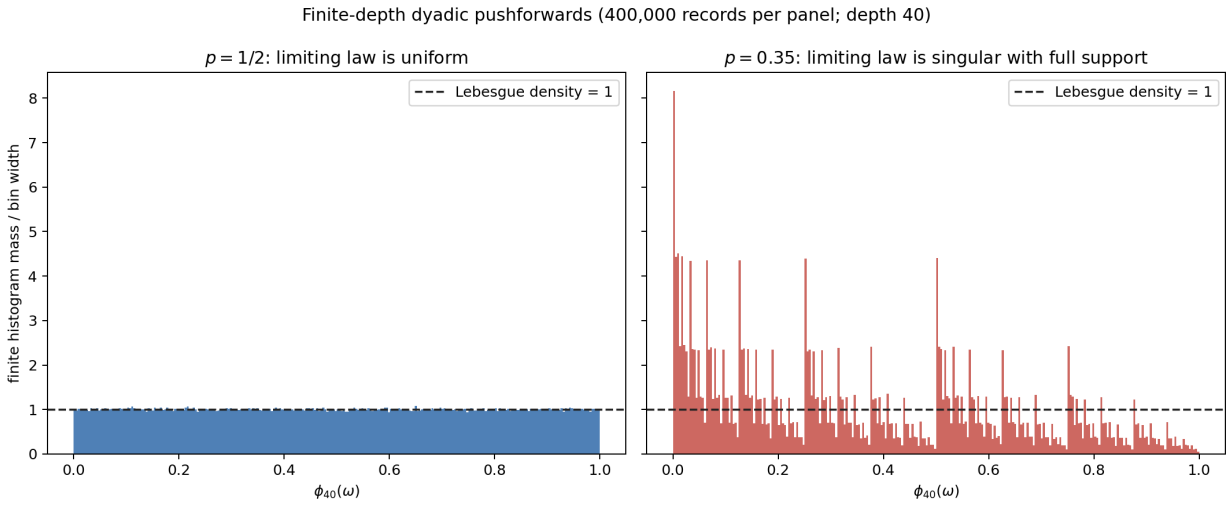


Figure 1: Finite samples under the dyadic map. The left histogram is the direct finite-resolution image of the fair pushforward. The right histogram shows unequal cylinder masses for $p = 0.35$. The analytic singularity statement does not depend on either histogram.

The left panel is visually flat because equal depth-8 cylinder masses coincide with the 256 equal histogram bins. The right panel exposes the multiplicative variation among those cylinder masses. Its structure is therefore not simulation noise and not a second proof of singularity; it is the expected finite-depth projection of the biased product law.

The dyadic calculation directly visualizes the exact law-to-continuum correspondence in [1] without changing the status of the underlying theorem.

4 Deterministic Arithmetic Binary Records

4.1 Three records and three denominators

The arithmetic comparison converts three deterministic objects into signs.

1. **Liouville record.** For every integer $n \geq 1$,
$$X_n = \lambda(n) = (-1)^{\Omega(n)} \in \{-1, +1\}.$$

2. **Möbius record.** Integers for which $\mu(n) = 0$ are ineligible. On the ordered squarefree integers, the record is
$$X_j = \mu(n_j) \in \{-1, +1\}.$$

3. **Prime-race record.** On the ordered odd primes p_j,
$$X_j = \begin{cases} +1, & p_j \equiv 1 \pmod 4, \\ -1, & p_j \equiv 3 \pmod 4. \end{cases}$$

For any one of these eligible-trial records, define

$$H_T = \sum_{j=1}^{T} \mathbf{1}_{\{X_j=+1\}}, \qquad W_T = \sum_{j=1}^{T} X_j = 2H_T - T.$$

Then

$$\widehat{P}_T(+1) = \frac{H_T}{T} = \frac{1}{2} + \frac{W_T}{2T}.$$

This identity is bookkeeping, not a probabilistic model. In particular, the symbol $\widehat{P}_T$ denotes an empirical proportion. It does not assert that the deterministic signs are independent or sampled from a Bernoulli law.

The common arithmetic cutoff $n \leq 3{,}000{,}000$ produces different eligible-trial counts:

Table 2: Finite arithmetic-record summaries at the common arithmetic cutoff.

Record	T	H_T	H_T/T	W_T	Exact ties	Sign changes
Liouville, all integers	3,000,000	1,499,529	0.499843	-942	9	1
Möbius, squarefree integers	1,823,773	911,940	0.500029	107	5,104	2,588
Odd primes modulo 4	216,815	108,283	0.499426	-249	111	48

A *sign change* in the last column means a change between positive and negative nonzero values of W_t after exact zero ties are omitted. An exact tie is counted separately. This definition matches the computation and avoids calling every zero or every plotted downward spike a crossing.

4.2 Half-frequency and finite texture

All three final proportions lie close to $1/2$, but the table shows why that single summary cannot describe the finite record. The Möbius walk repeatedly reaches and crosses zero. The Liouville walk has only one sign change after zero ties are removed within the same arithmetic cutoff. The prime-race record has the smallest eligible sample and retains the directional texture associated with the classical Chebyshev-bias problem [6,7].

For an additional denominator check, the audit truncates every record after the first 216,815 eligible trials. At that common trial horizon, the observed head frequencies are 0.499269 for Liouville, 0.499458 for Möbius, and 0.499426 for the odd-prime record. This comparison equalizes record length but not arithmetic cutoff, so it is reported only as a diagnostic. Neither comparison makes the three arithmetic mechanisms interchangeable.

The dashed slope $-1/2$ line in Figure 2 is the familiar independent fair-walk scale translated to empirical frequency. It is not fitted to the data. A finite record can follow, cross, or depart from that guide without establishing an asymptotic exponent. Precise RH-equivalence language is intentionally withheld from this draft until its exact source is locally available and directly checked.

4.3 Running sums

The running-sum view makes the distinction between limiting proportion and finite-scale balance visible. Since

$$\left|\widehat{P}_T(+1) - \frac{1}{2}\right| = \frac{|W_T|}{2T},$$

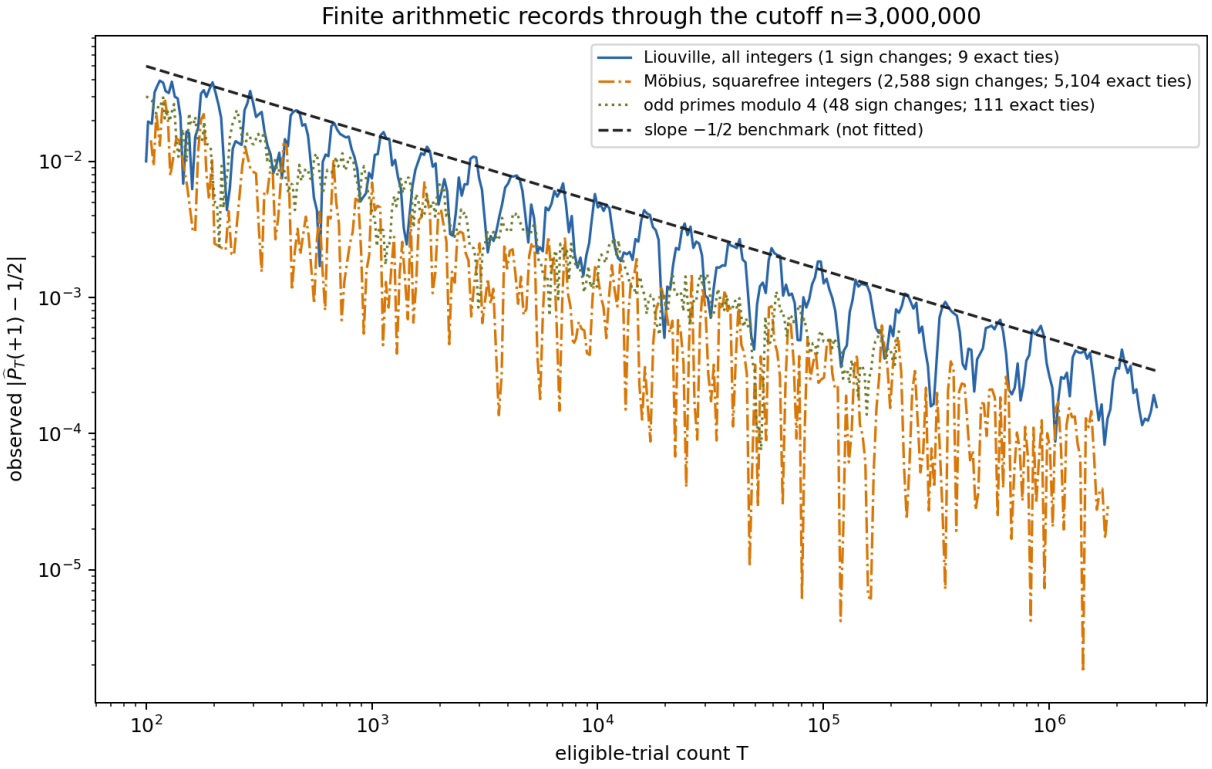


Figure 2: Observed absolute deviation from half-frequency against each record's eligible-trial count. The dashed line is a slope benchmark only. Zero-deviation sampled points cannot appear on the logarithmic axis; exact ties are counted from each full record in the legend. The legend's crossing count uses the sign-change definition given above.

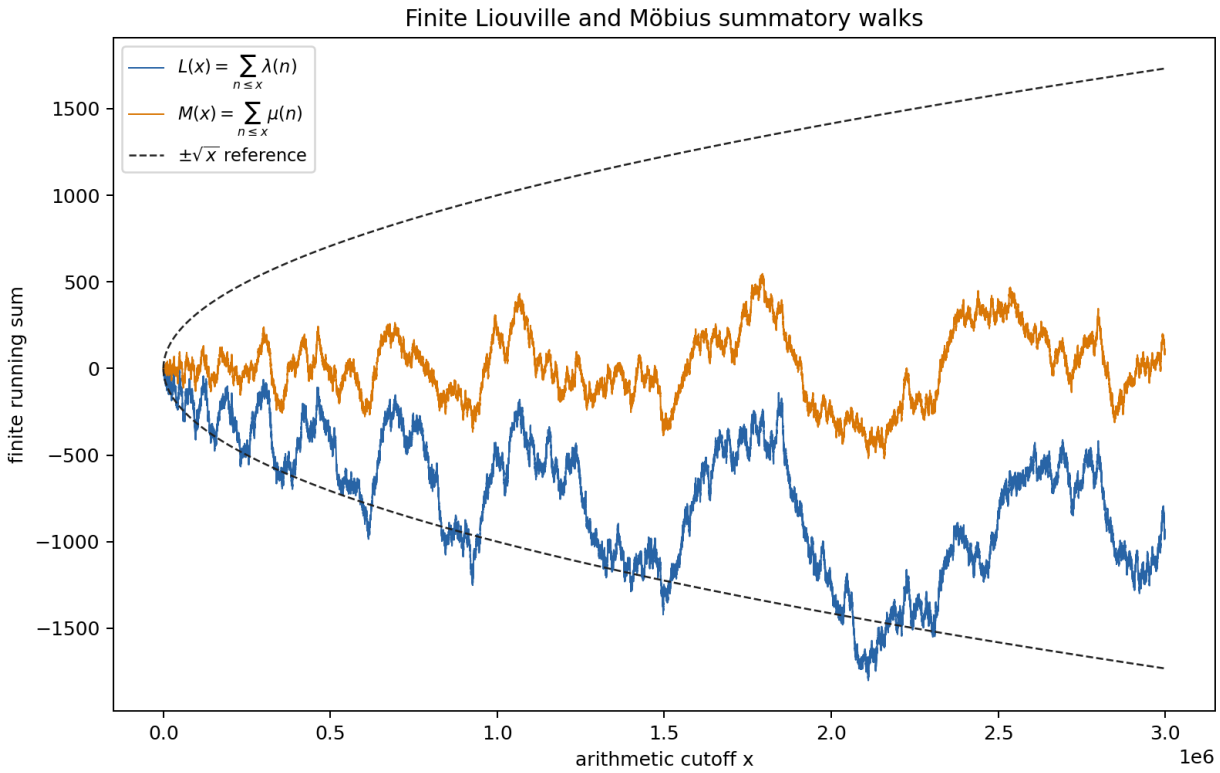


Figure 3: Liouville and Möbius summatory walks through $3{,}000{,}000$, shown with the descriptive envelope $\pm\sqrt{x}$. Here the horizontal axis is the arithmetic cutoff x, so the Möbius sum retains its zero terms. The envelope is a visual reference, not a proved bound for the displayed records.

sublinear walk growth is enough for the empirical proportion to approach $1/2$. It does not require the walk to alternate evenly around zero at finite scales. The strong one-sided appearance of the Liouville walk in the displayed interval therefore does not contradict its final near-half proportion. Likewise, the many exact ties of the squarefree Möbius record do not imply independent fair-coin behavior.

These comparisons provide a concrete warning: a half-frequency statement and a statement of finite-scale structural absence are different claims. The bare Bernoulli law in [1] includes both exchange symmetry and independent identical repetition. Neither assumption is imported into these arithmetic records.

5 Zeta-Zero Spectral-Statistics Comparison

5.1 Data and unfolding

The zero cache contains the first 1,200 positive ordinates t_n of nontrivial zeros of the Riemann zeta function, from $t_1 \approx 14.134725$ through $t_{1200} \approx 1648.270598$. The script removes ordinates at or below 50, where the smooth asymptotic density is least accurate, leaving 1,190 ordinates and 1,189 nearest-neighbor spacings.

The smooth Riemann–von Mangoldt term is used to unfold the ordinates:

$$U(t) = \frac{t}{2\pi} \log\left(\frac{t}{2\pi e}\right).$$

The nearest-neighbor spacings of $U(t_n)$ are compared with the GUE Wigner-surmise density

$$P_{\mathrm{GUE}}(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}$$

and the Poisson density e^{-s}. Pairwise differences up to three mean spacings are also histogrammed against Montgomery's limiting comparison

$$R_2(x) = 1 - \left(\frac{\sin \pi x}{\pi x}\right)^2.$$

5.2 Interpretive scope of the spectral comparison

The displayed suppression near zero spacing is qualitatively consistent with level repulsion rather than an uncorrelated Poisson picture. This is an established context, not a new result [4,5]. The computation is valuable here because the stated method reproduces the expected visual distinction on known data.

The spectral comparison is also where the notation $1/2$ is most likely to create a false connection. Three mathematically different uses must remain separate:

- $1/2$ as a probability in the bare binary law;
- $1/2$ as the exponent in a square-root fluctuation benchmark; and
- $1/2$ as the real-coordinate location of the zeta critical line.

The common numeral does not prove an identity among these objects. The zeta zeros are not binary outcomes, and the biconditional in [1] does not apply to them literally. Recent experimental work on repulsion near the central point of families of L-functions supplies a useful model for tightly scoped numerical research [8], but it does not convert this calibration into support for a new claim about the bare binary law.

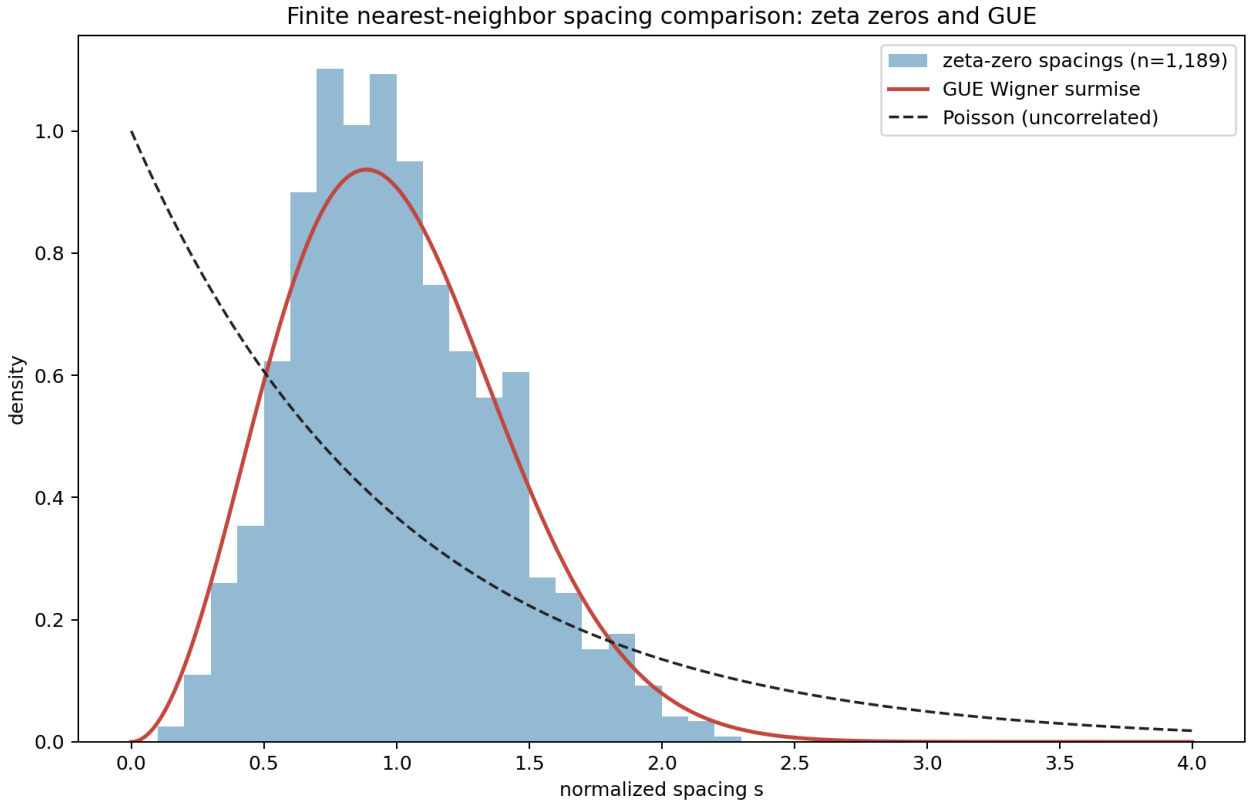


Figure 4: Nearest-neighbor spacings after smooth unfolding, compared with GUE and Poisson curves. The finite low-zero sample is a qualitative reproduction, not a new test of the Montgomery–Odlyzko phenomenon [4,5].

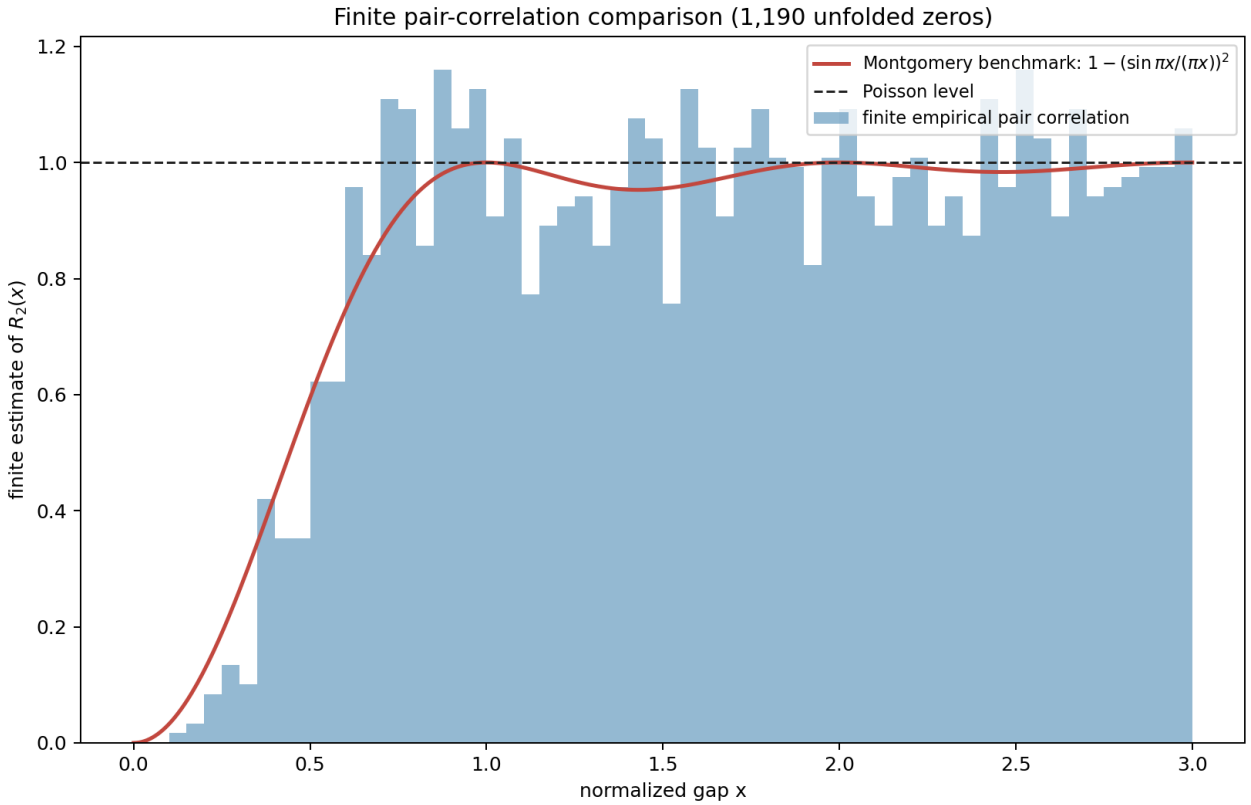


Figure 5: Finite empirical pair correlation from the same retained zeros, compared with Montgomery's limiting curve. Edge effects, the low height, smooth unfolding, and histogram choices limit quantitative interpretation.

6 Integrated Interpretation of the Three Computations

The complete computational sequence can now be read without collapsing its levels.

1. **The dyadic pushforward visualizes an exact component of [1].** Equal Bernoulli cylinder weights map to equal dyadic interval lengths; the finite histogram makes that correspondence visible.
2. **The arithmetic calculations test descriptive questions on deterministic binary records.** Near-half frequency can coexist with distinct tie counts, sign changes, directional texture, and eligibility rules. The resulting records are comparisons with the fair model, not instances assumed to satisfy it.
3. **The zeta-zero calculation reproduces a known spectral-statistics comparison.** It supplies wider number-theoretic and quantum-chaos context while remaining outside the proof chain.

The completed computations are internally interpretable within their respective scopes because their denominators, finite ties, sign-change definitions, and inferential distances are stated explicitly.

7 Limitations

The dyadic histogram is a finite Monte Carlo display of an analytic result. Changing its depth, sample size, or binning changes the plotted texture but not the theorem.

The arithmetic records are deterministic and finite. No independence test, asymptotic exponent estimate, or number-theoretic proof is supplied. The three records also use different eligibility rules, so a shared arithmetic cutoff does not imply a shared number of trials.

The spectral comparison uses only low zeros and the smooth leading unfolding term. It includes no goodness-of-fit statistic, uncertainty interval, robustness study over unfolding choices, or high-zero dataset. Its role is calibration. A publishable new experimental-number-theory result would require a new target, larger controlled data, robustness checks, and a precise conjecture not supplied by this reproduction.

The computations do not revise or correct the result in [1]. They make the already separate logical and empirical layers easier to inspect.

8 Connection to Observation-Record Convergence

A separate analysis concerns how an accumulating observation record can drive a conditional weight toward a definite binary endpoint and how that classical convergence structure compares with established repeated QND measurement results. Because that question introduces filtrations, martingales, identifiability, pointer weights, and a distinct quantum-convergence conjecture, it is developed independently rather than appended here.

9 Conclusion

Taken together, the three analyses complete the half-probability-convergence program related to *The Bare Binary Biconditional*. The dyadic calculation directly visualizes the fair pushforward and the contrasting biased product measure. The arithmetic calculations record near-half empirical proportions in three deterministic sequences while preserving their unequal denominators and sharply different finite textures. The spectral

calculation reproduces the established GUE comparison for a finite cache of zeta zeros and serves as a calibration rather than a premise.

The arithmetic-sign and zeta-zero studies were first generated as candidate research options alongside multiple other study options, independently of *The Bare Binary Biconditional*. Because the biconditional had already been formulated, the author subsequently selected those two studies for the half-probability-convergence program specifically because of their relationship to the biconditional. The established literature then allowed the synthesis paper [1] to be completed before the computational program. The dyadic study was added afterward as a direct computational representation of [1], and the present paper completes the resulting three-study program.

Their inclusion does not treat a shared numeral as proof or assign every occurrence of $1/2$ the same mathematical meaning. The biconditional's support comes from the established evidence chain synthesized in [1]; these computations examine the expected appearance of probability $1/2$ where a binary-probability interpretation applies, while making explicit the boundaries beyond which it does not. Together, the three completed studies close the loop of the proposed half-probability-convergence program while retaining distinct mathematical roles.

The first study therefore has a direct substantive relationship to the biconditional. The second also has a substantive role: it demonstrates that deterministic records can approach half-frequency without becoming Bernoulli or independent, directly clarifying what the biconditional does and does not establish. The third has a more limited connection and is explicitly presented as spectral calibration and broader research-program context, not as evidence for the biconditional.

The resulting contribution is computational and methodological: one exact measure identity, one deterministic arithmetic comparison, and one spectral calibration are presented together with their logical roles stated explicitly. The figures make the distinction between analytic identity, finite observation, and contextual comparison directly inspectable. They do not replace the theorem, turn deterministic records into Bernoulli trials, or turn a shared occurrence of $1/2$ into a mathematical equivalence.

References

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[2] É. Borel, "Les probabilités dénombrables et leurs applications arithmétiques," *Rendiconti del Circolo Matematico di Palermo* **27**, 247–271 (1909). https://doi.org/10.1007/BF03019651

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[8] Nicolás Coloma, Maria Espericueta Sandoval, Erika Lopez, Francisco Ponce, Gustavo Rama, Nathan C. Ryan, and Alejandro Vargas-Altamirano, "Repulsion of Zeros Close to $s = 1/2$ for L-Functions," arXiv:2401.07959v1 (2024). https://arxiv.org/abs/2401.07959v1

Source-Access Note

The full text of Borel's 1909 paper was not available for direct inspection in the source collection used for this project. It is cited only for the well-known normal-number attribution already bounded in [1]; no page-specific quotation or additional claim about the paper is made here. Exact local-source status for every reference is recorded in SOURCE_STATUS.md.

Preparation Note

Fable 5 proposed an initial set of experimental study options. The author selected the zeta-zero/GUE and deterministic half-frequency studies from among the options connected to probability convergence because he connected them to his pre-formulated bare binary biconditional, and he then directed the resulting research program. The dyadic-pushforward visualization was subsequently added from the completed biconditional paper while completing the experimental program that would have preceded that paper had the biconditional not already been supported by pre-existing literature. The present follow-up completes that program in support of the now-extant paper. GPT 5.6 assisted with source audit, computation audit, figure preparation, and drafting. The author retains responsibility for the claims, interpretations, and final text.